EXPERIMENT – 26

Implementation of RSA Public-Key Encryption Scheme

CODE:

```
#include <stdio.h>

int power(int a, int b, int n)
{
    long r = 1;
    while (b--)
        r = (r * a) % n;
    return r;
}

int main()
{
    int p = 17, q = 11;
    int n = p * q, phi = (p - 1) * (q - 1);
    int e = 7, d = 23;
    int m = 88, c, decrypted;

    c = power(m, e, n);
    decrypted = power(c, d, n);

    printf("Public Key  : (%d, %d)\n", e, n);
    printf("Private Key : (%d, %d)\n", d, n);
    printf("Plaintext   : %d\n", m);
    printf("Ciphertext  : %d\n", c);
```

```
    printf("Decrypted   : %d\n", decrypted);

    return 0;
}
```

OUTPUT: